MedVision AI Knowledge Base
Pleural Effusion

1. Overview

Pleural effusion is the abnormal accumulation of fluid within the pleural space between the visceral and parietal pleura. It is a manifestation of an underlying disease rather than a disease itself. Small effusions may be asymptomatic, while large effusions can significantly impair breathing by compressing the lung.

2. Causes

Common causes include congestive heart failure, pneumonia (parapneumonic effusion), malignancy, tuberculosis, pulmonary embolism, liver cirrhosis, kidney disease, trauma, autoimmune disorders, and postoperative complications. Effusions are broadly classified as transudative or exudative based on their underlying mechanism.

3. Symptoms

Patients may present with shortness of breath, pleuritic chest pain, dry cough, reduced exercise tolerance, fatigue, and in severe cases respiratory distress. Symptoms often depend on the volume of accumulated fluid and the underlying disease.

4. Risk Factors

Risk factors include heart failure, chronic kidney disease, liver disease, pneumonia, lung cancer, smoking, tuberculosis exposure, pulmonary embolism, autoimmune diseases, recent thoracic surgery, and chest trauma.

5. Chest X-ray Findings

Typical radiographic findings include blunting of the costophrenic angle, homogeneous lower lung opacity, meniscus sign, obscuration of the hemidiaphragm, mediastinal shift with very large effusions, and associated pneumonia or lung collapse in some patients. Lateral decubitus views and ultrasound improve detection of small effusions.

6. Diagnosis

Diagnosis combines history, physical examination, chest X-ray, lung ultrasound, CT chest when indicated, thoracentesis, pleural fluid analysis using Light's criteria, microbiology, cytology, and laboratory investigations directed at the underlying cause.

7. Differential Diagnosis

Conditions that may mimic pleural effusion include pulmonary edema, lower lobe atelectasis, elevated diaphragm, pleural thickening, lung mass, diaphragmatic eventration, and extensive pneumonia without pleural fluid.

8. Severity Classification

Effusions may be classified as small, moderate, or large based on imaging. Clinically they are also categorized as uncomplicated parapneumonic effusion, complicated parapneumonic effusion, empyema, malignant pleural effusion, or transudative effusion.

9. Treatment

Management targets the underlying cause. Diuretics are commonly used for heart failure-related effusions. Antibiotics are indicated for infectious causes. Therapeutic thoracentesis relieves symptoms in large effusions. Recurrent malignant effusions may require indwelling pleural catheters or pleurodesis.

10. Complications

Potential complications include respiratory failure, empyema, fibrothorax, trapped lung, recurrent effusions, infection following procedures, and reduced quality of life.

11. Prevention

Preventive strategies focus on controlling underlying diseases, prompt treatment of pneumonia, smoking cessation, vaccination where appropriate, regular follow-up for chronic illnesses, and early medical evaluation of persistent breathlessness.

12. Prognosis

Prognosis depends primarily on the underlying cause. Heart failure-related effusions often improve with medical treatment, whereas malignant pleural effusions may recur and indicate advanced disease. Early diagnosis generally improves outcomes.

13. Patient Explanation

Pleural effusion means extra fluid has collected around your lung rather than inside it. This fluid limits normal lung expansion and can make breathing difficult. Treatment aims to remove the fluid when necessary and address the condition that caused it.

14. AI Interpretation

AI detection of pleural effusion should be considered decision support. Review the highlighted region, estimate the size of the effusion, correlate imaging with symptoms and clinical findings, and confirm the diagnosis with a qualified radiologist or physician.

15. Quick Facts

Organ: Pleural space
Imaging: Chest X-ray
Typical Finding: Blunted costophrenic angle and meniscus sign
Common Symptom: Shortness of breath
Potential Emergency: Large effusion causing respiratory compromise
Clinical Correlation Required: Yes

16. References

World Health Organization (WHO); MedlinePlus (U.S. National Library of Medicine); Merck Manual Professional Edition; British Thoracic Society Pleural Disease Guidelines; American College of Chest Physicians (CHEST) guidance; standard radiology textbooks and open educational radiology resources.